

Complex Analysis – 8

Гармонические функции и интеграл Пуассона

1 сентября 2026 года

Теоретический минимум.

Опр. u is *harmonic* if $u \in C^2$ and $\Delta u = u_{xx} + u_{yy} = 0$. Locally $u = \operatorname{Re} f$ for a holomorphic f , and the *harmonic conjugate* v is determined up to a constant by the Cauchy–Riemann equations.

Теорема 1 (среднее и принцип максимума) $u(a) = \frac{1}{2\pi} \int_0^{2\pi} u(a + re^{i\theta}) d\theta$ for every disc contained in the domain; consequently a harmonic function on a domain attains neither an interior maximum nor an interior minimum unless it is constant.

Теорема 2 (интеграл Пуассона) For continuous boundary data φ on $\partial\mathbb{D}$ the function

$$u(re^{i\theta}) = \frac{1}{2\pi} \int_0^{2\pi} \frac{1-r^2}{1-2r\cos(\theta-t)+r^2} \varphi(t) dt$$

is harmonic in $\mathbb{D}$, continuous up to the boundary and equal to φ there; it is the unique such function.

Факт 1 Неравенство Харнака: for a positive harmonic u on $\mathbb{D}$, $\frac{1-|z|}{1+|z|}u(0) \leq u(z) \leq \frac{1+|z|}{1-|z|}u(0)$; отсюда принцип Харнака о монотонных пределах и теоремы типа Лиувилля.

1. Prove the mean value property and the maximum principle for harmonic functions, show that a harmonic function on a simply connected domain has a harmonic conjugate, and give an example on the annulus where it does not.
2. (S.T. Yau CSMC 2014) Let U be a bounded function on $\mathbb{R}$ with finitely many discontinuities. Prove that $P_U(x+iy) = \frac{1}{\pi} \int_{\mathbb{R}} \frac{y}{(x-\xi)^2+y^2} U(\xi) d\xi$

is harmonic in the upper half-plane and converges to $U(\xi_0)$ as $z \rightarrow \xi_0$ at every point of continuity of U .

3. (S.T. Yau CSMC 2013) Let u be a positive harmonic function on the punctured plane $\mathbb{C} \setminus \{0\}$. Show that u is constant.
4. (S.T. Yau CSMC 2023) Prove that a bounded harmonic function $u(x, y)$ on the domain $\{y > x^2\}$ with boundary values $u(x, x^2) = 0$ is identically zero.
5. (S.T. Yau CSMC 2017) Let E be a compact subset of a domain Ω . Prove that there is a constant $M > 0$, depending only on E and Ω , such that every positive harmonic function u on Ω satisfies $u(z_2) \leq Mu(z_1)$ for all $z_1, z_2 \in E$.